Quantization of the electromagnetic field

Week

VI

Until now, we have treated the atom quantum mechanically and the electromagnetic field classically. In the semiclassical picture, the field drives transitions, creates atomic coherences, and induces a dipole moment. The expectation value of this dipole determines the polarization that enters Maxwell's equations. We have shown that it explains a plethora of optical resonant optical phenomena, from Rabi oscillations to saturated absorption, AC Stark shift, and electromagnetically induced transparency.

However, so far, we have not described the quantum state of the radiation itself. In particular, it cannot distinguish in the classical picture states with distinct statistics (e.g. a field containing exactly one photon from a field whose photon number fluctuates). We shall now quantize the electromagnetic field, following first a cavity scenario, before entering multimode free space description [1]. We will show that the amplitude obeys the equation of a harmonic oscillator, and thus apply the quantum-mechanical rules to perform canonical quantization, interpreting its excitations as photons. After extending the construction to several modes and polarizations, we will introduce number states, coherent states, and squeezed states. Finally, we use the density operator to describe thermal radiation.

[1] *C. C. Gerry and P. L. Knight,* Introductory Quantum Optics*, Cambridge (2005).*

1 A cavity mode as a harmonic oscillator

We will first look at the case of an empty cavity, for which we assume a length L along z and transverse area S, with volume $V = SL$. We also assume a one-dimensional model, meaning that the selected mode is uniform across the transverse area, polarized along x, and reflected by perfectly conducting endplates at $z = 0, L$. Thus, we can write

$$\mathbf{E} = E_x(z,t)\mathbf{e}_x, \qquad \mathbf{B} = B_y(z,t)\mathbf{e}_y. \tag{1.1}$$

As there are no charges or currents in the interior, Maxwell's equations then reduce to

$$\frac{\partial E_x}{\partial z} = -\frac{\partial B_y}{\partial t}, \tag{1.2}$$

$$-\frac{\partial B_y}{\partial z} = \frac{1}{c^2}\frac{\partial E_x}{\partial t}, \qquad c^2 = \frac{1}{\epsilon_0 \mu_0}. \tag{1.3}$$

It is easy to take the time/spatial derivative of each equation and obtain a harmonic oscillator equation for each field. In particular, as the tangential electric field vanishes at each conducting endplate, a standing-wave pattern takes the form

$$E_x(z,t) = C\,Q(t)\sin(kz), \qquad k = \frac{m\pi}{L}, \quad m = 1, 2, \ldots, \tag{1.4}$$

with m labeling the *mode* and $Q(t)$ containing the dynamics. Substitution into Eq. (1.3) gives the oscillating part of the magnetic field as

$$B_y(z,t) = \frac{C}{c^2 k}\dot{Q}(t)\cos(kz). \tag{1.5}$$

Further substitution into Faraday's law then gives

$$CkQ\cos(kz) = -\frac{C}{c^2 k}\ddot{Q}\cos(kz), \tag{1.6}$$

so that we end up with the harmonic oscillator equation as previously said

$$\boxed{\ddot{Q} + \omega^2 Q = 0, \qquad \omega = ck.} \tag{1.7}$$

1.1 Energy and normalization

The electromagnetic energy stored in the cavity is given by the classical Hamiltonian,

$$H = \frac{1}{2}\int_V \left(\epsilon_0 E_x^2 + \frac{B_y^2}{\mu_0}\right) d^3 r, \tag{1.8}$$

and using the fact that

$$\int_V \sin^2(kz)\, d^3 r = \int_V \cos^2(kz)\, d^3 r = \frac{V}{2}, \tag{1.9}$$

we obtain

$$H = \frac{\epsilon_0 V C^2}{4} Q^2 + \frac{V C^2}{4\mu_0 c^4 k^2}\dot{Q}^2. \tag{1.10}$$

We now take a clever choice of $C = \sqrt{2\omega^2/(\epsilon_0 V)}$ and define $P = \dot{Q}$. The field and its energy then become

$$E_x(z,t) = \sqrt{\frac{2\omega^2}{\epsilon_0 V}}\, Q(t)\sin(kz), \tag{1.11}$$

$$B_y(z,t) = \sqrt{\frac{2\mu_0}{V}}\, P(t)\cos(kz), \tag{1.12}$$

and thus

$$\boxed{H = \frac{1}{2}P^2 + \frac{1}{2}\omega^2 Q^2.} \tag{1.13}$$

The electric field contributes $\omega^2 Q^2/2$ and the magnetic field contributes $P^2/2$. Note that interestingly, energy is exchanged between these contributions during an oscillation[2], while their sum remains constant. Hamilton's equations reproduce Maxwell's dynamics:

$$\dot{Q} = \frac{\partial H}{\partial P} = P, \qquad \dot{P} = -\frac{\partial H}{\partial Q} = -\omega^2 Q. \tag{1.14}$$

1.2 Canonical quantization

To perform quantization, we can simply replace the classical variables by Hermitian operators and impose the canonical commutator,

$$Q \longrightarrow \hat{Q}, \qquad P \longrightarrow \hat{P}, \qquad [\hat{Q}, \hat{P}] = i\hbar. \tag{1.15}$$

The Hamiltonian is now

$$\hat{H} = \frac{1}{2}\hat{P}^2 + \frac{1}{2}\omega^2\hat{Q}^2. \tag{1.16}$$

Rather than solve a differential equation in a coordinate representation, it is convenient to introduce the dimensionless operators

$$\boxed{\hat{a} = \frac{\omega\hat{Q} + i\hat{P}}{\sqrt{2\hbar\omega}}, \qquad \hat{a}^\dagger = \frac{\omega\hat{Q} - i\hat{P}}{\sqrt{2\hbar\omega}}.} \tag{1.17}$$

Their inverses are

$$\hat{Q} = \sqrt{\frac{\hbar}{2\omega}}(\hat{a} + \hat{a}^\dagger), \qquad \hat{P} = -i\sqrt{\frac{\hbar\omega}{2}}(\hat{a} - \hat{a}^\dagger). \tag{1.18}$$

[2] *note however that Q and P are* mass-normalized mode variables, *not the position and momentum of a material particle. In particular, Q does not have the units of a length:* $[Q] = \mathrm{J}^{1/2}\mathrm{s}$ *and* $[P] = \mathrm{J}^{1/2}$.

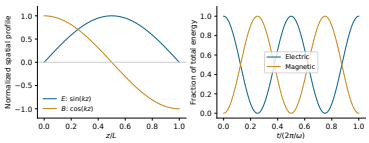


Figure 1. A single cavity mode. The electric and magnetic spatial profiles are displaced by a quarter wavelength for this polarization. The panel below shows the exchange of electric and magnetic energy.

Using $[\hat{Q}, \hat{P}] = i\hbar$ gives

$$[\hat{a}, \hat{a}^\dagger] = \frac{-i\omega[\hat{Q}, \hat{P}] + i\omega[\hat{P}, \hat{Q}]}{2\hbar\omega} = 1, \tag{1.19}$$

$$\hat{a}^\dagger\hat{a} = \frac{\omega^2\hat{Q}^2 + \hat{P}^2 - \hbar\omega}{2\hbar\omega}. \tag{1.20}$$

Consequently[3],

$$\boxed{\hat{H} = \hbar\omega\left(\hat{n} + \frac{1}{2}\right), \qquad \hat{n} = \hat{a}^\dagger\hat{a}.} \tag{1.21}$$

[3] *Note that the order of the operators matters:* $\hat{a}\hat{a}^\dagger = \hat{n} + 1$, *whereas* $\hat{a}^\dagger\hat{a} = \hat{n}$.

1.3 The ladder and the number states

In the form presented in equation 1.21 we should now try to find the normalized eigenstates of $\hat{n}$. Suppose $|\nu\rangle$ is a normalized with eigenvalue ν. The commutators

$$[\hat{n}, \hat{a}] = -\hat{a}, \qquad [\hat{n}, \hat{a}^\dagger] = \hat{a}^\dagger \tag{1.22}$$

imply directly that[4]

$$\hat{n}(\hat{a}|\nu\rangle) = (\nu - 1)(\hat{a}|\nu\rangle), \tag{1.23}$$

$$\hat{n}(\hat{a}^\dagger|\nu\rangle) = (\nu + 1)(\hat{a}^\dagger|\nu\rangle), \tag{1.24}$$

[4] *show the commutators and these relations is a good exercise...*

which means that $\hat{a}^\dagger$ raises the eigenvalue by 1 and $\hat{a}$ lowers the eigenvalue by 1. But repeated lowering must terminate, since a nonzero state cannot have a negative eigenvalue. The lowest state therefore shall satisfy

$$\hat{a}|0\rangle = 0. \tag{1.25}$$

Starting with this state, repeated raising gives the nonnegative integer spectrum $n = 0, 1, 2, \ldots$. [5]

Definition. Photon-number states, or Fock states - For a single mode, $|n\rangle$ is an eigenstate of $\hat{n}$ that is constructed as

$$|n\rangle = \frac{(\hat{a}^\dagger)^n}{\sqrt{n!}}|0\rangle. \tag{1.27}$$

and satisfies

$$\hat{n}|n\rangle = n|n\rangle, \qquad E_n = \hbar\omega\left(n + \frac{1}{2}\right), \tag{1.28}$$

$$\hat{a}|n\rangle = \sqrt{n}\,|n-1\rangle, \qquad \hat{a}^\dagger|n\rangle = \sqrt{n+1}\,|n+1\rangle \tag{1.29}$$

The energy difference between adjacent states is $\hbar\omega$.

[5] *The normalization factors follow from*

$$\|\hat{a}|n\rangle\|^2 = n, \|\hat{a}^\dagger|n\rangle\|^2 = n + 1. \tag{1.26}$$

To an excitation of the mode (+1) we call it a **photon**. The labels n count excitations of one fixed mode. Note that the state $|0\rangle$ is the **vacuum** of that mode, and not the zero vector of the Hilbert space. Indeed it is normalized and has energy $\hbar\omega/2$.

By hermiticity of $\hat{n}$, number states form an orthonormal basis,

$$\langle m|n\rangle = \delta_{mn}, \qquad \sum_{n=0}^{\infty} |n\rangle\langle n| = \hat{I}, \tag{1.30}$$

and a general pure state can be constructed as

$$|\psi\rangle = \sum_{n=0} c_n |n\rangle, \qquad \sum_n |c_n|^2 = 1. \tag{1.31}$$

For calculations, the matrix elements are particularly useful:

$$\langle m|\hat{a}|n\rangle = \sqrt{n}\,\delta_{m,n-1}, \qquad \langle m|\hat{a}^\dagger|n\rangle = \sqrt{n+1}\,\delta_{m,n+1}. \tag{1.32}$$

The operators $\hat{Q}$, $\hat{P}$, $\hat{n}$ and $\hat{H}$ are Hermitian observables. The operators $\hat{a}$ and $\hat{a}^\dagger$ are not Hermitian, yet their Hermitian combinations are important as they are related with measurable field amplitudes. Indeed, substituting Eq. (1.18) into Eqs. (1.11)–(1.12) gives

$$\hat{E}_x(z) = \mathcal{E}_{\rm vac}(\hat{a} + \hat{a}^\dagger)\sin(kz), \tag{1.33}$$

$$\hat{B}_y(z) = -\frac{i\mathcal{E}_{\rm vac}}{c}(\hat{a} - \hat{a}^\dagger)\cos(kz), \tag{1.34}$$

with $\mathcal{E}_{\rm vac} = \sqrt{\frac{\hbar\omega}{\epsilon_0 V}}$.

1.4 Time evolution of states or operators

AS we already discussed, in the **Schrödinger picture** the state evolves and the operators above have no explicit time dependence. Since the Hamiltonian is diagonal in the number basis, we have

$$|\psi(t)\rangle = \sum_n c_n(0) e^{-i\omega(n+1/2)t}|n\rangle. \tag{1.35}$$

A number state acquires only an overall phase, while a superposition develops relative phases.

In the **Heisenberg picture**, the state is fixed and the operators evolve under

$$\frac{d\hat{a}_H}{dt} = \frac{i}{\hbar}[\hat{H}, \hat{a}_H] = -i\omega\hat{a}_H. \tag{1.36}$$

Therefore

$$\hat{a}_H(t) = \hat{a}e^{-i\omega t}, \qquad \hat{a}_H^\dagger(t) = \hat{a}^\dagger e^{i\omega t}, \tag{1.37}$$

where the unlabelled operators on the right are evaluated at $t = 0$. The electric field then becomes

$$\hat{E}_x(z,t) = \mathcal{E}_{\text{vac}} \sin(kz) \left(\hat{a} e^{-i\omega t} + \hat{a}^\dagger e^{i\omega t} \right). \tag{1.38}$$

Either picture gives the same expectation value. It is typically useful to separate the positive and negative-frequency parts as

$$\hat{E}^{(+)} = \mathcal{E}_{\text{vac}} \sin(kz) \hat{a} e^{-i\omega t}, \qquad \hat{E}^{(-)} = (\hat{E}^{(+)})^\dagger. \tag{1.39}$$

The name positive frequency refers to the Fourier convention $e^{-i\omega t}$ with $\omega > 0$, and in this convention, the positive-frequency field contains annihilation operators.

1.5 Quadratures and the uncertainty relation

A classical monochromatic wave has two independent real amplitudes multiplying $\cos \omega t$ and $\sin \omega t$. Their quantum counterparts are related with the dimensionless **quadrature operators**,

$$\hat{X}_1 = \frac{\hat{a} + \hat{a}^\dagger}{2}, \qquad \hat{X}_2 = \frac{\hat{a} - \hat{a}^\dagger}{2i}, \qquad \hat{a} = \hat{X}_1 + i\hat{X}_2. \tag{1.40}$$

as it can be easily shown that (1.38) can be written as

$$\hat{E}_x(z,t) = 2\mathcal{E}_{\text{vac}} \sin(kz) \left(\hat{X}_1 \cos \omega t + \hat{X}_2 \sin \omega t \right). \tag{1.41}$$

The commutator for this set of operators is

$$[\hat{X}_1, \hat{X}_2] = \frac{i}{2}, \tag{1.42}$$

and according to the Heisenberg uncertainty principle[6]

$$\text{Var}(\hat{X}) \, \text{Var}(\hat{Y}) \geq \frac{1}{4} |\langle [\hat{X}, \hat{Y}] \rangle|^2 \tag{1.43}$$

then we have

$$\boxed{\text{Var}(\hat{X}_1) \, \text{Var}(\hat{X}_2) \geq \frac{1}{16}.} \tag{1.44}$$

[6] $\text{Var}(\hat{O}) = \langle \hat{O}^2 \rangle - \langle \hat{O} \rangle^2$

2 Beyond cavity: generalization for several modes

The single-mode approximation is equivalent to one oscillator only. In general the classical field is a sum of orthogonal normal modes that do not interact,

which means that their orthogonality removes the cross terms in the integrated field energy, leaving the Hamiltonian simply as

$$H = \sum_j \frac{1}{2}\left(P_j^2 + \omega_j^2 Q_j^2\right). \tag{2.1}$$

In this case we can still apply canonical quantization with

$$[\hat{Q}_j, \hat{P}_l] = i\hbar\delta_{jl}, \qquad [\hat{Q}_j, \hat{Q}_l] = [\hat{P}_j, \hat{P}_l] = 0, \tag{2.2}$$

and defining a ladder operator for each mode gives

$$[\hat{a}_j, \hat{a}_l^\dagger] = \delta_{jl}, \qquad [\hat{a}_j, \hat{a}_l] = 0, \tag{2.3}$$

$$\hat{H} = \sum_j \hbar\omega_j \left(\hat{a}_j^\dagger \hat{a}_j + \frac{1}{2}\right). \tag{2.4}$$

Different modes thus correspond to independent quantum degrees of freedomm a their joint number basis is a tensor-product basis,

$$|n_1, n_2, \ldots\rangle = |n_1\rangle \otimes |n_2\rangle \otimes \cdots . \tag{2.5}$$

For example, $|1,0\rangle$ and $|0,1\rangle$ both contain one photon, but in different modes.

2.1 Plane waves and two transverse polarizations

Rather than quantizing the electric and magnetic fields independently, we will first express both fields in terms of electromagnetic potentials. Since

$$\boldsymbol{\nabla} \cdot \mathbf{B} = 0, \tag{2.6}$$

we introduce a vector potential $\mathbf{A}$ such that

$$\boxed{\mathbf{B} = \boldsymbol{\nabla} \times \mathbf{A}.} \tag{2.7}$$

Faraday's law then gives

$$\boldsymbol{\nabla} \times \mathbf{E} = -\frac{\partial}{\partial t}\left(\boldsymbol{\nabla} \times \mathbf{A}\right) \Rightarrow \boldsymbol{\nabla} \times \left(\mathbf{E} + \frac{\partial \mathbf{A}}{\partial t}\right) = 0 \tag{2.8}$$

and consequently, we can introduce a scalar potential ϕ and write

$$\boxed{\mathbf{E} = -\boldsymbol{\nabla}\phi - \frac{\partial \mathbf{A}}{\partial t}.} \tag{2.9}$$

We now note that these two potentials are sufficient to construct both electric and magnetic field, yet they are not unique. For any sufficiently

regular scalar function $\chi(\mathbf{r}, t)$, the transformation

$$\mathbf{A}' = \mathbf{A} + \boldsymbol{\nabla}\chi, \qquad \phi' = \phi - \frac{\partial \chi}{\partial t} \tag{2.10}$$

leaves both $\mathbf{E}$ and $\mathbf{B}$ unchanged. Indeed, the curl of a gradient vanishes, and the two additional contributions to the electric field cancel. This freedom to change the potentials without changing the physical fields is called **gauge freedom**. For the free radiation field, a convenient choice is the *Coulomb gauge*,

$$\boldsymbol{\nabla} \cdot \mathbf{A} = 0. \tag{2.11}$$

In the absence of charges, Gauss's law becomes

$$0 = \boldsymbol{\nabla} \cdot \mathbf{E} = -\nabla^2 \phi - \frac{\partial}{\partial t} (\boldsymbol{\nabla} \cdot \mathbf{A}) = -\nabla^2 \phi. \tag{2.12}$$

With periodic boundary conditions and no imposed electrostatic field, ϕ is spatially constant. Its remaining time dependence can be removed by a gauge transformation. We may therefore set $\phi = 0$, obtaining

$$\boxed{\mathbf{E} = -\frac{\partial \mathbf{A}}{\partial t}, \qquad \mathbf{B} = \boldsymbol{\nabla} \times \mathbf{A},} \tag{2.13}$$

which means that $\boldsymbol{A}$ alone is sufficient to describe the electromagnetic field in free space. It is trivial to show that the vector potential also follows a wave equation under this gauge choice,

$$\boxed{\nabla^2 \mathbf{A} - \frac{1}{c^2} \frac{\partial^2 \mathbf{A}}{\partial t^2} = 0.} \tag{2.14}$$

For a plane wave, the condition $\boldsymbol{\nabla} \cdot \mathbf{A} = 0$ becomes $\mathbf{k} \cdot \mathbf{e}_{\mathbf{k}\lambda} = 0$, which means that **two independent polarization components remain for each propagation direction** and that should also be considered. The classical vector potential can then be expanded in traveling wave modes,

$$\mathbf{A}(\mathbf{r}, t) = \sum_{\mathbf{k},\lambda} \left[\mathbf{e}_{\mathbf{k}\lambda} A_{\mathbf{k}\lambda} e^{i\Phi_{\mathbf{k}}} + \text{c.c.} \right], \qquad \Phi_{\mathbf{k}} = \mathbf{k} \cdot \mathbf{r} - \omega_{\mathbf{k}} t. \tag{2.15}$$

Here $A_{\mathbf{k}\lambda}$ is a constant complex amplitude that specifies the magnitude and phase of the mode, and c.c. denotes the complex conjugate of the preceding

term. Differentiating gives

$$\mathbf{E}(\mathbf{r},t) = i\sum_{\mathbf{k},\lambda}\omega_{\mathbf{k}}\left[\mathbf{e}_{\mathbf{k}\lambda}A_{\mathbf{k}\lambda}e^{i\Phi_{\mathbf{k}}} - \mathbf{e}^{*}_{\mathbf{k}\lambda}A^{*}_{\mathbf{k}\lambda}e^{-i\Phi_{\mathbf{k}}}\right], \tag{2.16}$$

$$\mathbf{B}(\mathbf{r},t) = i\sum_{\mathbf{k},\lambda}\left[(\mathbf{k}\times\mathbf{e}_{\mathbf{k}\lambda})A_{\mathbf{k}\lambda}e^{i\Phi_{\mathbf{k}}} - (\mathbf{k}\times\mathbf{e}^{*}_{\mathbf{k}\lambda})A^{*}_{\mathbf{k}\lambda}e^{-i\Phi_{\mathbf{k}}}\right]. \tag{2.17}$$

Finally, to determine the normalization of these amplitudes, we calculate again the the electromagnetic energy,

$$H = \frac{1}{2}\int_V\left(\epsilon_0\mathbf{E}^2 + \frac{\mathbf{B}^2}{\mu_0}\right)d^3r. \tag{2.18}$$

Using the orthogonality relation

$$\int_V e^{i(\mathbf{k}-\mathbf{k}')\cdot\mathbf{r}}\,d^3r = V\delta_{\mathbf{k}\mathbf{k}'}, \tag{2.19}$$

together with polarization orthogonality, gives

$$\boxed{H = 2\epsilon_0 V\sum_{\mathbf{k},\lambda}\omega^2_{\mathbf{k}}|A_{\mathbf{k}\lambda}|^2.} \tag{2.20}$$

A clever choice of dimensionless complex amplitudes $\alpha_{\mathbf{k}\lambda}$ through

$$A_{\mathbf{k}\lambda} = \sqrt{\frac{\hbar}{2\epsilon_0\omega_{\mathbf{k}}V}}\,\alpha_{\mathbf{k}\lambda}, \tag{2.21}$$

the energy takes a simple and convenient form

$$H = \sum_{\mathbf{k},\lambda}\hbar\omega_{\mathbf{k}}|\alpha_{\mathbf{k}\lambda}|^2. \tag{2.22}$$

With this normalization, the classical fields are

$$\mathbf{A}(\mathbf{r},t) = \sum_{\mathbf{k},\lambda}\sqrt{\frac{\hbar}{2\epsilon_0\omega_{\mathbf{k}}V}}\left[\mathbf{e}_{\mathbf{k}\lambda}\alpha_{\mathbf{k}\lambda}e^{i\Phi_{\mathbf{k}}} + \text{c.c.}\right], \tag{2.23}$$

$$\mathbf{E}(\mathbf{r},t) = i\sum_{\mathbf{k},\lambda}\sqrt{\frac{\hbar\omega_{\mathbf{k}}}{2\epsilon_0 V}}\left[\mathbf{e}_{\mathbf{k}\lambda}\alpha_{\mathbf{k}\lambda}e^{i\Phi_{\mathbf{k}}} - \text{c.c.}\right], \tag{2.24}$$

$$\mathbf{B}(\mathbf{r},t) = i\sum_{\mathbf{k},\lambda}\sqrt{\frac{\hbar}{2\epsilon_0\omega_{\mathbf{k}}V}}\left[(\mathbf{k}\times\mathbf{e}_{\mathbf{k}\lambda})\alpha_{\mathbf{k}\lambda}e^{i\Phi_{\mathbf{k}}} - \text{c.c.}\right]. \tag{2.25}$$

Second quantization: promoting amplitudes to operators

We have already established that each electromagnetic normal mode has the dynamics of a harmonic oscillator. With this knowledge, and instead of

performing first quantization in canonical variables, we can now apply the quantization rule derived previously directly to its dimensionless amplitude, i.e.

$$\boxed{\alpha_{\mathbf{k}\lambda} \longrightarrow \hat{a}_{\mathbf{k}\lambda}, \qquad \alpha^*_{\mathbf{k}\lambda} \longrightarrow \hat{a}^\dagger_{\mathbf{k}\lambda},} \tag{2.26}$$

with the operators satisfying the usual bosonic commutation relations

$$[\hat{a}_{\mathbf{k}\lambda}, \hat{a}^\dagger_{\mathbf{k}'\lambda'}] = \delta_{\mathbf{k}\mathbf{k}'}\delta_{\lambda\lambda'}, \tag{2.27}$$

$$[\hat{a}_{\mathbf{k}\lambda}, \hat{a}_{\mathbf{k}'\lambda'}] = [\hat{a}^\dagger_{\mathbf{k}\lambda}, \hat{a}^\dagger_{\mathbf{k}'\lambda'}] = 0. \tag{2.28}$$

[7]

[7] *annihilation and creation operators belonging to different modes commute.*

The resulting vector-potential operator is

$$\boxed{\hat{\mathbf{A}}(\mathbf{r}, t) = \sum_{\mathbf{k},\lambda} \sqrt{\frac{\hbar}{2\epsilon_0 \omega_{\mathbf{k}} V}} \left[\mathbf{e}_{\mathbf{k}\lambda} \hat{a}_{\mathbf{k}\lambda} e^{i\Phi_{\mathbf{k}}} + \text{H.c.}\right].} \tag{2.29}$$

and the electric and magnetic field operators are

$$\hat{\mathbf{E}}(\mathbf{r}, t) = i \sum_{\mathbf{k},\lambda} \sqrt{\frac{\hbar\omega_{\mathbf{k}}}{2\epsilon_0 V}} \left[\mathbf{e}_{\mathbf{k}\lambda} \hat{a}_{\mathbf{k}\lambda} e^{i\Phi_{\mathbf{k}}} - \text{H.c.}\right], \tag{2.30}$$

$$\hat{\mathbf{B}}(\mathbf{r}, t) = i \sum_{\mathbf{k},\lambda} \sqrt{\frac{\hbar}{2\epsilon_0 \omega_{\mathbf{k}} V}} \left[(\mathbf{k} \times \mathbf{e}_{\mathbf{k}\lambda}) \hat{a}_{\mathbf{k}\lambda} e^{i\Phi_{\mathbf{k}}} - \text{H.c.}\right]. \tag{2.31}$$

Here H.c. denotes the Hermitian conjugate of the preceding term. Note that all three field operators are Hermitian. Note also that these expressions use the Heisenberg picture for the free field: the operators $\hat{a}_{\mathbf{k}\lambda}$ on the right are evaluated at $t = 0$, while their free evolution is included in $e^{-i\omega_{\mathbf{k}} t}$.

Finally, inserting the quantized fields into the electromagnetic energy gives

$$\hat{H} = \sum_{\mathbf{k},\lambda} \frac{\hbar\omega_{\mathbf{k}}}{2} \left(\hat{a}^\dagger_{\mathbf{k}\lambda} \hat{a}_{\mathbf{k}\lambda} + \hat{a}_{\mathbf{k}\lambda} \hat{a}^\dagger_{\mathbf{k}\lambda}\right) \tag{2.32}$$

$$= \boxed{\sum_{\mathbf{k},\lambda} \hbar\omega_{\mathbf{k}} \left(\hat{n}_{\mathbf{k}\lambda} + \frac{1}{2}\right)}, \qquad \hat{n}_{\mathbf{k}\lambda} = \hat{a}^\dagger_{\mathbf{k}\lambda} \hat{a}_{\mathbf{k}\lambda}. \tag{2.33}$$

The symmetric operator combination follows from the quadratic field energy. It retains the vacuum contribution, which would be missed by simply replacing $|\alpha_{\mathbf{k}\lambda}|^2$ with $\hat{a}^\dagger_{\mathbf{k}\lambda} \hat{a}_{\mathbf{k}\lambda}$ in the already simplified classical energy.

Polarization is a mode degree of freedom - For a fixed wavevector along z, take horizontal and vertical linear polarizations with operators

$\hat{a}_H, \hat{a}_V$. The one-photon subspace is spanned by

$$|1_H, 0_V\rangle, \qquad |0_H, 1_V\rangle. \tag{2.34}$$

A photon in a definite superposition of polarizations is described by

$$|\psi\rangle = c_H|1_H, 0_V\rangle + c_V|0_H, 1_V\rangle, \qquad |c_H|^2 + |c_V|^2 = 1. \tag{2.35}$$

Equivalently,

$$|\psi\rangle = \hat{b}^\dagger|0_H, 0_V\rangle, \qquad \hat{b}^\dagger = c_H\hat{a}_H^\dagger + c_V\hat{a}_V^\dagger. \tag{2.36}$$

Then $[\hat{b}, \hat{b}^\dagger] = 1$. Real equal coefficients describe a linear polarization at 45°; equal magnitudes with relative phase $\pm\pi/2$ describe the two circular polarizations.